Problems with original designs

01

Column-wise loading

Reconsidered when we needed to assemble four output pixels into one complete 32-bit write.

02

Large register file

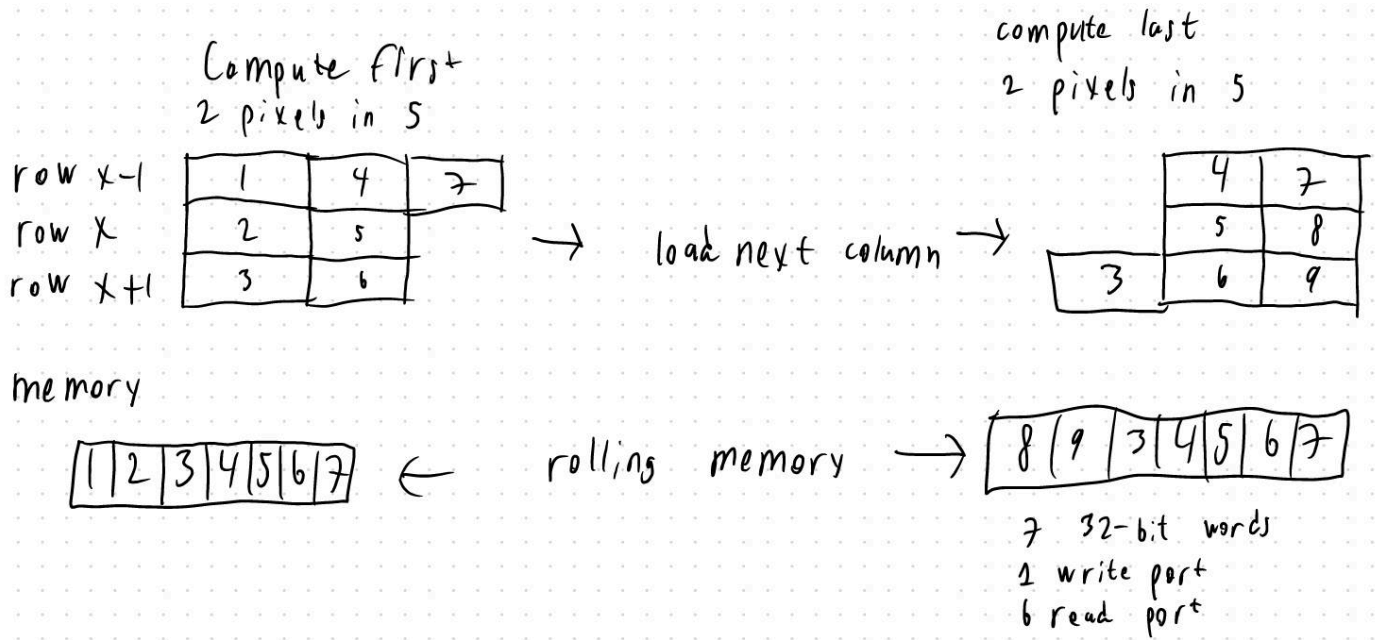
About 600 words and nine read ports. A straightforward read-mux estimate alone used **≈88% of available LUTs**.

03

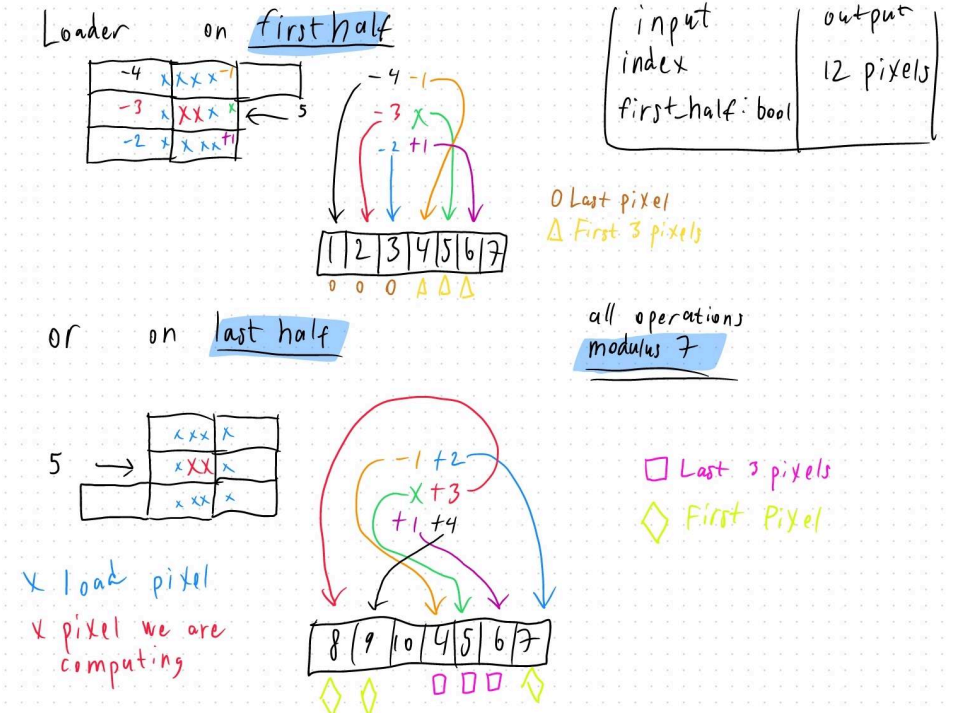
Develop a working version

Keep it simple and maybe slighlty less efficient. Then we can consider redesigning it

Compute 1 row at a time and keep a rolling window of 7 words in memory



Selecting the pixel neighbourhoods



Status and Implementation Details

Implemented

- Memory
- Loader
- Parts of the FSM

> Today's goal: a running accelerator.

Processing schedule

- Two pixels computed in parallel
- Each input word read three times
- One word read per cycle

≈1,315 frames/s

25,344 words/frame = 288 × 352 / 4

100,000,000 / (25,344 × 3) ≈ 1,315

Optimistic estimate at clock speed of 100MHz

Finish, integrate, then optimize

01 **Integrate with the SoC**

Connect and verify the complete accelerator flow.

02 **Improve buffering**

Reuse data so each input word is read fewer times.

03 **Add resource sharing**

Reduce duplicated hardware where the schedule allows.

04 **Support border pixels**

Extend processing to the image boundaries.